



ESTIMAÇÃO E CONTROLO PREDITIVO DISTRIBUÍDO

PROJECT REPORT

From System Identification to Model Predictive Control: Application to a Thermal Process

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The group of students identified above guarantees that the text of this report and all the software and results delivered were entirely carried out by the elements of the group, with a significant participation of all of them, and that no part of the work or the software and results presented was obtained from other people or sources.

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P1 – Basics on Constrained Optimization

The objective of this part was to solve a two-dimensional optimization problem using MATLAB Optimization Toolbox. The Rosenbrock function was minimized from the initial estimate

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \quad x_0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Two cases were considered: the unconstrained problem, solved with `fminunc`, and the constrained problem, solved with `fmincon`, subject to

$$x_1 \leq 0.5, \quad Ax \leq b, \quad A = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad b = 0.5.$$

For the unconstrained case, the optimizer converged to the global minimum of the Rosenbrock function:

$$x^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad f(x^*) = 0.$$

However, this point violates the constraint $x_1 \leq 0.5$. Therefore, the constrained optimum must lie on the boundary $x_1 = 0.5$. Along this boundary, the Rosenbrock valley is minimized for $x_2 = x_1^2 = 0.25$, which gives

$$x_c^* = \begin{bmatrix} 0.5 \\ 0.25 \end{bmatrix}, \quad f(x_c^*) = 0.25.$$

Table 1: Optimization results for the Rosenbrock function.

Case	Optimum point	Objective value
Unconstrained	$x^* = (1, 1)$	0
Constrained, $x_1 \leq 0.5$	$x_c^* = (0.5, 0.25)$	0.25

Figure 1 shows the surface plot of the Rosenbrock function together with the constraint plane $x_1 = 0.5$. The narrow curved valley is visible, which explains the numerical difficulty of the problem. Since the unconstrained minimum is located at $(1, 1)$, it lies outside the feasible region.

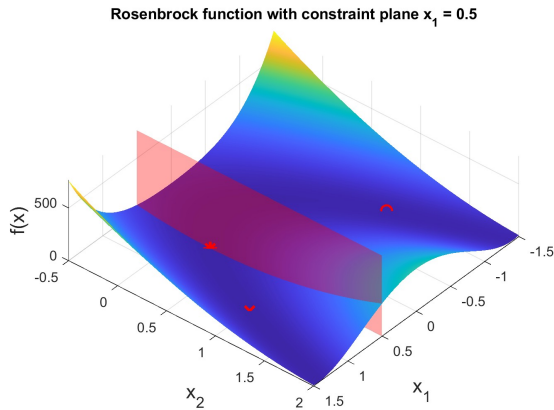


Figure 1: Surface plot of the Rosenbrock function with the constraint plane $x_1 = 0.5$.

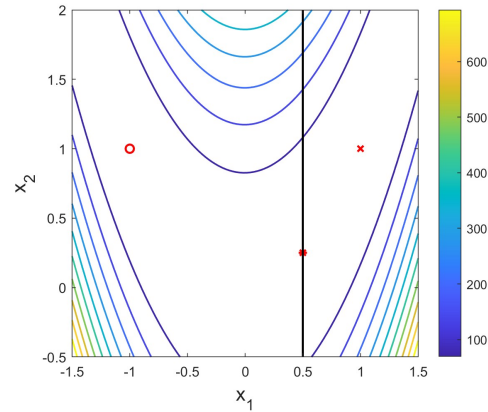


Figure 2: Contour plot of the Rosenbrock function, showing the initial estimate, the unconstrained minimum and the constrained minimum.

Figure 2 shows the contour plot requested in the work statement. The initial estimate is marked with ‘o’, the unconstrained minimum with ‘x’, and the constrained minimum with ‘*’. The vertical line represents the constraint boundary $x_1 = 0.5$, with the feasible region on its left side.

The results show that constraints can change both the location and the value of the optimum. Without constraints, the global minimum is reached at $(1, 1)$, with zero cost. With the constraint $x_1 \leq 0.5$, this point is no longer admissible and the best feasible solution becomes $(0.5, 0.25)$, with cost 0.25. This illustrates the importance of constrained optimization in Model Predictive Control, where physical limits on inputs, states and outputs must be explicitly respected.

P2 – Basics on Receding Horizon Control

The objective of this part was to compare finite-horizon receding-horizon control with the infinite-horizon Linear Quadratic regulator. Two first-order discrete-time plants were analysed:

$$x(t+1) = Ax(t) + u(t),$$

with $B = 1$, $C = 1$, and $Q = 1$. The first plant is open-loop unstable, with $A = 1.2$, and the second is open-loop stable, with $A = 0.8$.

The infinite-horizon LQ gain was computed using

$$[K_{LQ}, S, \lambda] = \text{dlqr}(A, B, Q, R).$$

The receding-horizon gain K_{RH} was then computed for different prediction horizons H . The selected control penalties were

$$R \in \{0.1, 1, 10, 100\},$$

covering both aggressive controllers, for small R , and conservative controllers, for large R . The horizon was tested up to $H = 30$, which was sufficient to observe the convergence of K_{RH} to K_{LQ} .

Table 2: Infinite-horizon LQ gains for both plants.

Plant	R	K_{LQ}	$ A - BK_{LQ} $
$A = 1.2$	0.1	1.1026	0.0974
$A = 1.2$	1	0.7935	0.4065
$A = 1.2$	10	0.4882	0.7118
$A = 1.2$	100	0.3844	0.8156
$A = 0.8$	0.1	0.7309	0.0691
$A = 0.8$	1	0.4624	0.3376
$A = 0.8$	10	0.1399	0.6601
$A = 0.8$	100	0.0207	0.7793

Unstable plant: $A = 1.2$

Figure 3 shows the evolution of the receding-horizon gain and the magnitude of the closed-loop eigenvalue for the unstable plant. The upper plot compares K_{RH} with the corresponding K_{LQ} , while the lower plot compares $|\lambda_{cl}|$ with the stability boundary $|\lambda| = 1$.

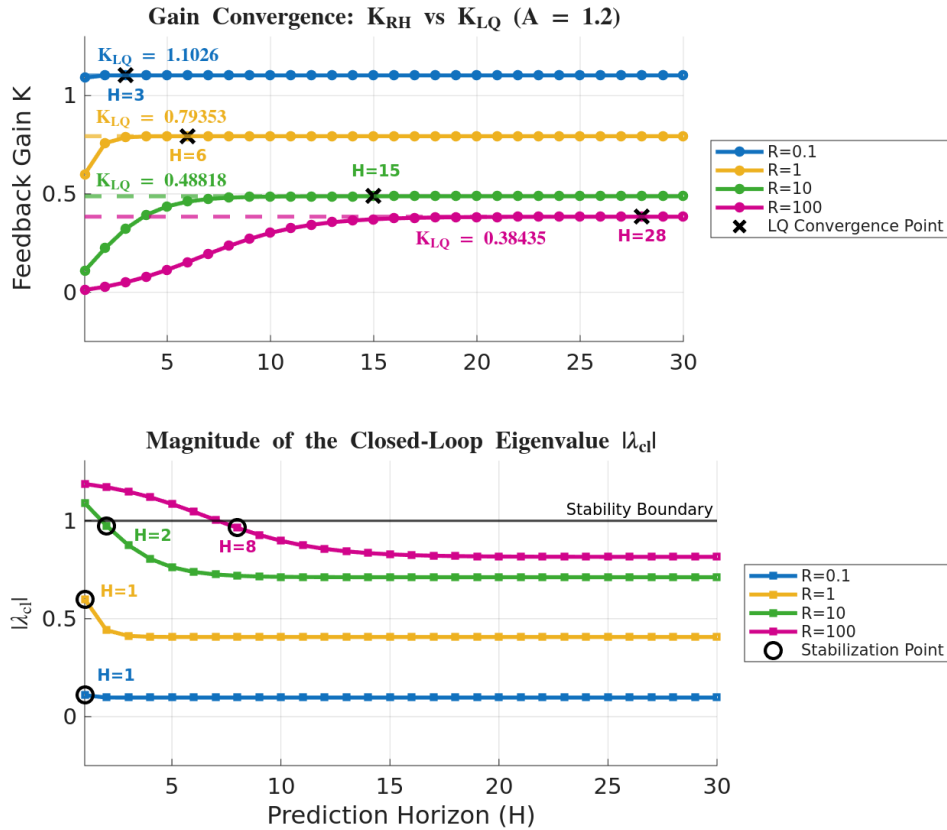


Figure 3: Gain convergence and closed-loop eigenvalue magnitude for the unstable plant $A = 1.2$.

For all tested values of R , K_{RH} increases with the prediction horizon and converges to the corresponding infinite-horizon gain. Smaller values of R produce larger gains because the control effort is weakly penalized, while larger R values lead to more conservative actions and require longer horizons to approach the LQ solution. The closed-loop eigenvalue is

$$\lambda_{cl} = A - BK_{RH}.$$

Since $A = 1.2$ is open-loop unstable, short horizons may be insufficient to stabilize the system, especially for large R . As H increases, the controller anticipates the future growth of the unstable state and brings $|\lambda_{cl}|$ below one. Thus, for this plant, enlarging the horizon is important both for performance and stability.

Stable plant: $A = 0.8$

The same study was repeated for the open-loop stable plant. Figure 4 shows the corresponding receding-horizon gain convergence and closed-loop eigenvalue magnitude.

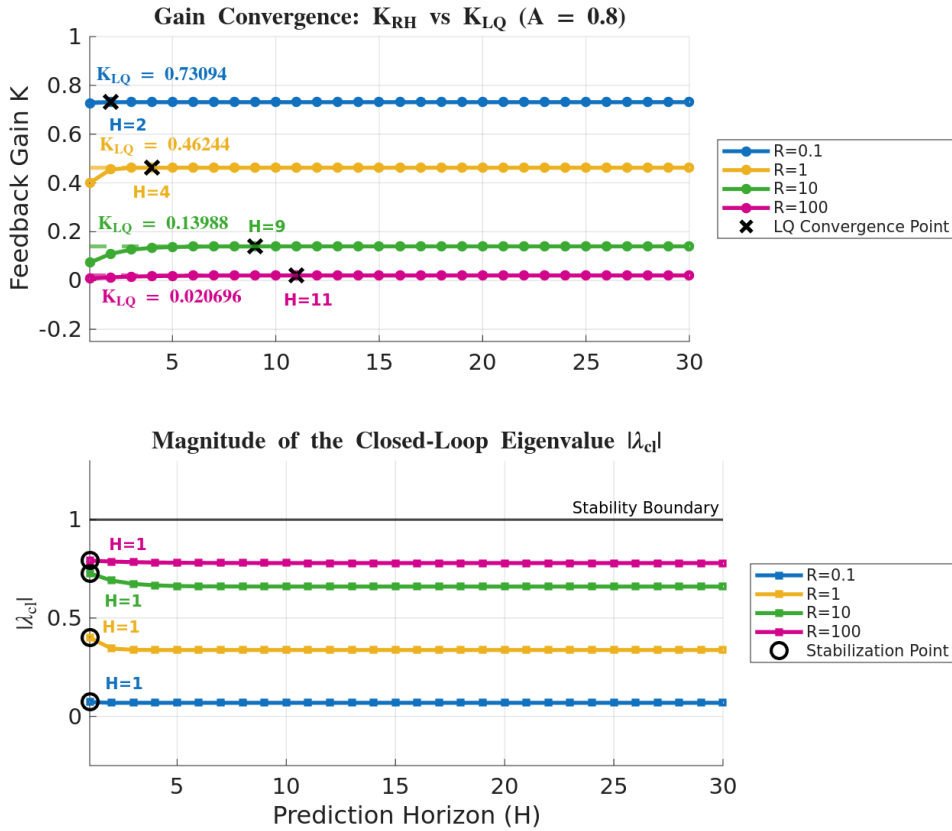


Figure 4: Gain convergence and closed-loop eigenvalue magnitude for the stable plant $A = 0.8$.

For this plant, the open-loop eigenvalue already satisfies

$$|A| = 0.8 < 1,$$

so the system is stable even without feedback. As a result, the closed-loop eigenvalue remains below the stability boundary for all tested values of R and H . As in the unstable case, K_{RH} converges to K_{LQ} as H increases, but the stability implications are less critical because the plant is already naturally stable.

Comparison and discussion

Increasing the prediction horizon has two main effects: it makes the receding-horizon gain approach the infinite-horizon LQ gain, and it can improve closed-loop stability by producing a more forward-looking control action. This effect is much more relevant for the unstable plant. When $A = 1.2$, a short horizon may not be enough to stabilize the system, particularly when R is large and the control action is heavily penalized. In this case, a larger H allows the controller to account for the future growth of the state and move the closed-loop eigenvalue inside the unit circle.

For the stable plant, $A = 0.8$, the system is already stable in open loop. Therefore, larger horizons mainly improve the approximation to the optimal LQ performance, but they are not essential for stability. In relation to

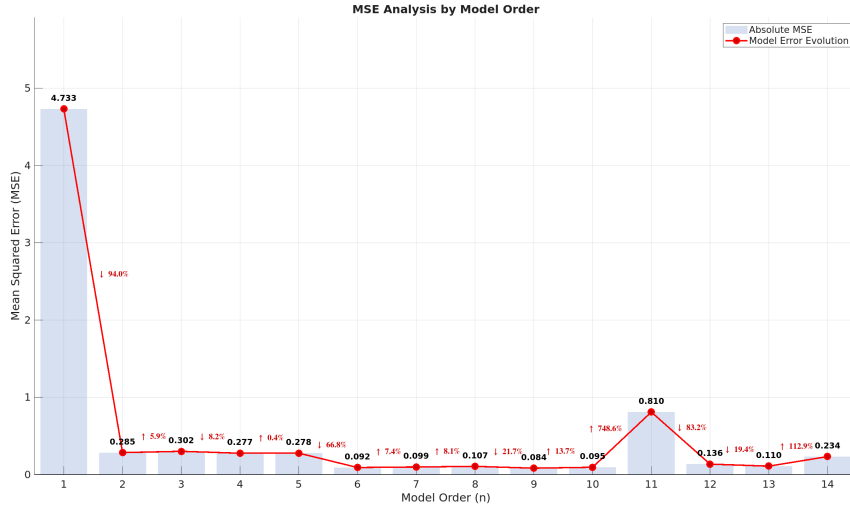


Figure 5: Validation MSE obtained for each tested model order.

stability, it is therefore more advantageous to use enlarged values of H for the unstable plant. Overall, the horizon should be selected as a compromise between performance, stability requirements and real-time computational cost.

P3 – Model Identification

1. Obtaining Training and Validation Data

Before collecting the experimental data, it was necessary to determine a suitable input value for Heater 1 that would drive the TCLab system to an equilibrium temperature close to 45°C. For this purpose, the provided TCLab_heaterSliders_test.slx Simulink model was used. An input of 25% was found to achieve the desired operating point at an equilibrium of 46.3°C. The data acquisition script was then modified to use a sampling time of $T_s = 5$ s.

The input profile was specified as a vector of step values, where each entry corresponded to a constant input level applied to Heater 1. For the identification experiment, the following training profile was used:

$$[25, 22.5, 20, 17.5, 27.5, 15, 30, 12.5]\%.$$

The first step lasted 2000 s, in order to ensure that the system reached the initial steady state. The remaining steps lasted 1000 s each, allowing the temperature to approach a new steady-state value after each input change. Equilibrium values were estimated using samples between 1000 and 2000 seconds of the initial constant-input interval.

A second experiment was performed for model validation. This data set was new to the identified model and it was used to evaluate whether the identified model could reproduce the dynamics of the system under a different input sequence.

The validation input profile was more aggressive than the training profile, with shorter step durations and larger input variations. The profile used was

$$[5, 85, 5, 20, 45, 80, 15, 0, 15, 75, 30, 15, 90, 25, 5, 20, 10, 55, 80, 60]\%.$$

Each step lasted 50 s, resulting in a total experiment duration of 1000 s. Since the step duration was short, the system did not reach steady state during each interval.

2. Model Identification and Order Selection

The main design choice in the identification procedure was the model order, represented by the number of states n . Several model orders were tested, ranging from $n = 1$ to $n = 14$. The resulting validation mean squared error (MSE) is presented in Figure 5.

As observed in Figure 5, a significant reduction in the MSE occurs when switching from first-order to second-order models, remaining consistent from the second to the tenth order with very little decrease. Beyond this point, the error increases, which could indicate model overfitting.

To make a well-founded decision, a more detailed analysis was conducted on the **second**, **sixth**, and **tenth** order models by comparing their outputs against the experimental data and their absolute errors. These three

specific architectures were selected because they represent critical inflection points in the MSE reduction trend discussed previously.

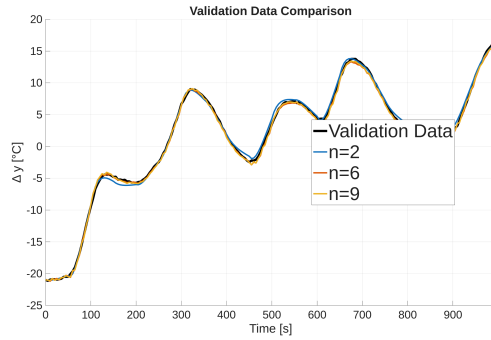


Figure 6: Validation comparison between the study models outputs against experimental data.

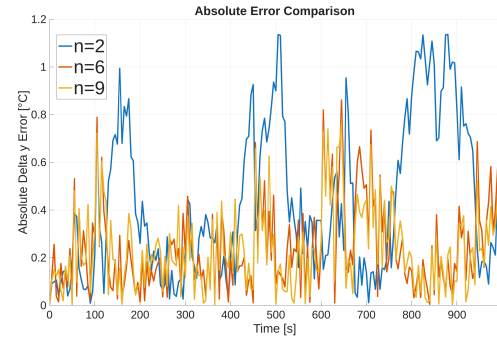


Figure 7: Absolute Error comparison between the study models.

As can be seen, the observed performance in Figures. 6 and 7 are very similar, especially in the absolute error. The difference between the second-order and tenth-order models is no greater than 1°C, besides the max error achieved by the the added complexity provides no significant benefit. Thus, the second-order model is the best choice.

3. Model Performance

After selecting the model order, the performance of the final model was evaluated both on the training dataset and on the validation dataset.

As expected identified model follows the training data output closely in Figure 8. Likewise even in the validation data it managed to follow meaning it grasped the dynamics well in Figure 9

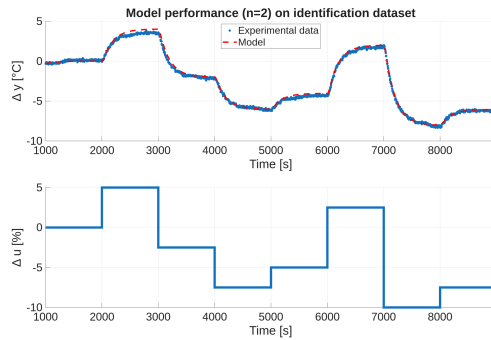


Figure 8: Model performance with $n = 2$ on the training data.

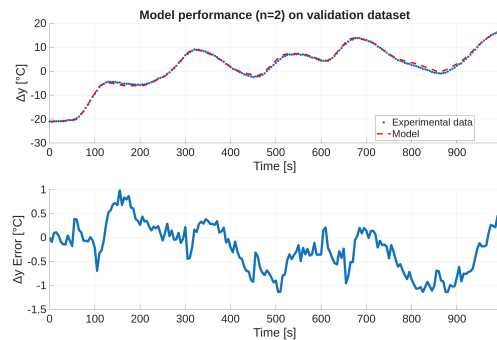


Figure 9: Model performance with $n = 2$ on the validation data.

4. Discussion

The identification procedure produced a discrete-time linear incremental model of the TCLab single-heater system, suitable for control design around the selected equilibrium point ($u_{ss} = 25\%$, $y_{ss} = 46.3^\circ\text{C}$). Multiple model orders were tested and evaluated based on validation MSE and error analysis.

The second-order model ($n = 2$) achieved the best performance/complexity trade-off, with a validation MSE = 0.285. Higher-order models failed to improve validation results significantly, indicating that increased complexity beyond $n = 2$ does not enhance generalization. This suggests that a low-order model sufficiently captures the system dynamics for the intended control application.

The final model was saved as `singleheater_model.mat` and will be used for simulation and MPC controller design in subsequent project stages.

P4 – MPC and Kalman Filter Design

This part of the project addresses the design of a Model Predictive Controller (MPC) for the TCLab single-heater system. The controller is built incrementally: first the unconstrained regulator is validated against the analytical

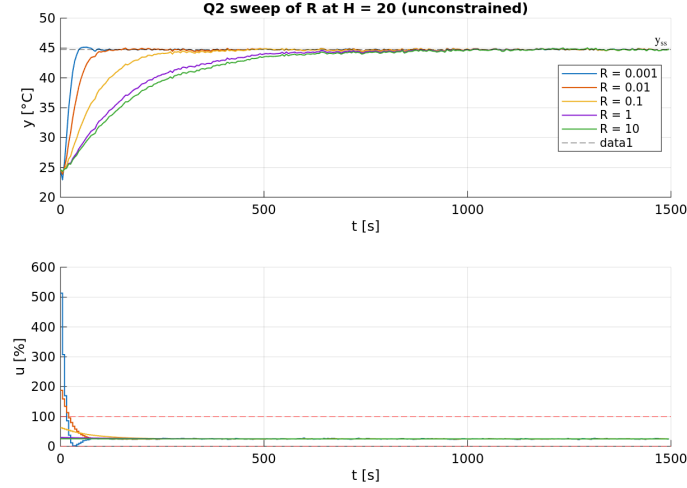


Figure 10: Closed-loop response for several values of R at $H = 20$, without constraints.

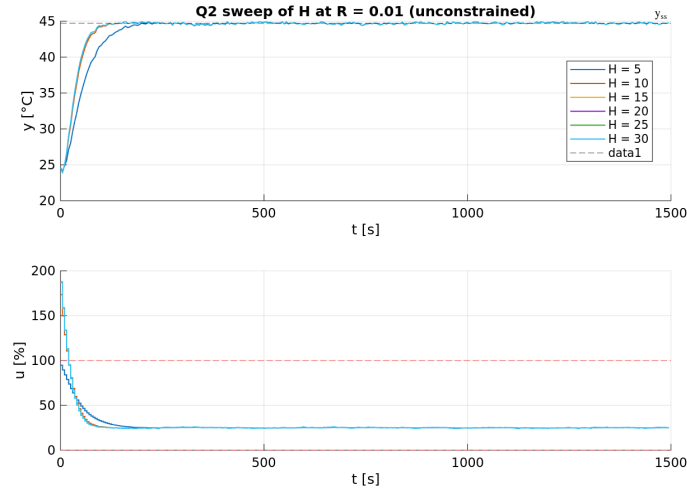


Figure 11: Closed-loop response for several values of H at $R = 0.01$. Performance saturates from $H = 15$ onwards.

receding-horizon solution of P2, then it is augmented with input bounds, reference tracking with feedforward, and a soft safety cap on the output. All experiments in this section are performed in simulation using the linear incremental model identified in P3.

2. Closed-Loop Unconstrained Regulator: Choice of H and R

The first closed-loop experiment uses the MPC as a regulator: Δy is driven to zero, so the absolute output is driven to y_{ss} . The simulation starts from the equilibrium for $u = 0$ (ambient temperature, $y(0) \approx 23^\circ\text{C}$) and lasts $T = 1500$ s. No constraints are active at this stage. The objective is to identify a horizon H and a control weight R to be fixed for the remainder of P4.

A grid sweep was performed over $H \in \{5, 10, 15, 20, 25, 30\}$ and $R \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$. The cost function used is

$$J = \sum_{i=0}^{H-1} \hat{y}(i+1)^2 + R \hat{u}(i)^2,$$

penalising deviations of the predicted output from the equilibrium and the magnitude of the control increment.

Figure 10 shows the effect of R for a fixed $H = 20$, and Figure 11 shows the effect of H for a fixed $R = 0.01$.

Three regimes are clearly visible. With $R = 10^{-3}$ the controller is very aggressive and would request $u \approx 550\%$ at the first sample. With $R = 10^{-2}$ the loop is fast and well damped, asking for $u \approx 215\%$ at the start. From

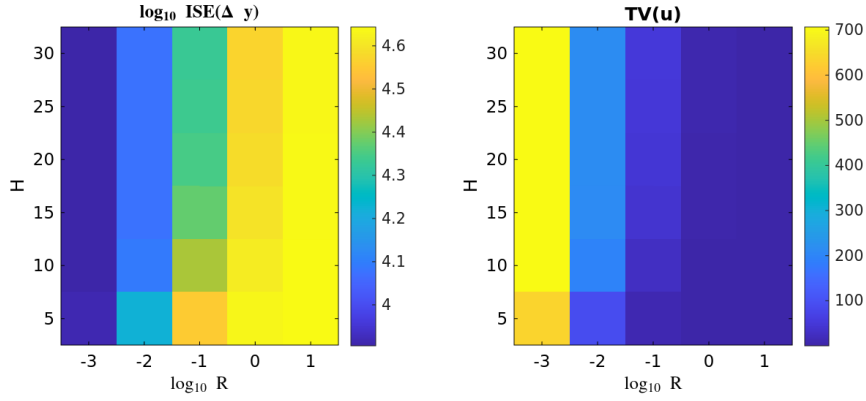


Figure 12: Summary of the $H \times R$ sweep. Left: $\log_{10}$ of the ISE on Δy . Right: total variation of the control signal.

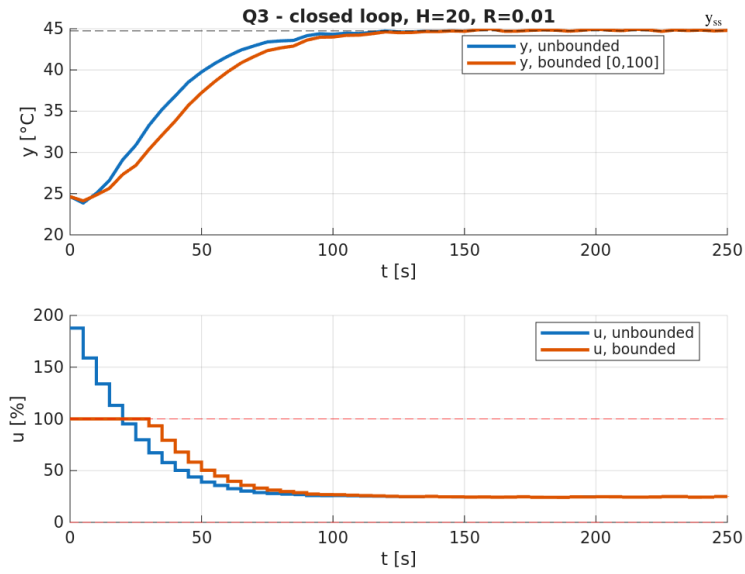


Figure 13: Closed-loop response with and without input bounds.

$R = 10^{-1}$ onwards the controller becomes overly conservative; $R = 10$ does not even reach y_{ss} within 1500 s. As for H , the curves overlap from $H = 15$, confirming that the receding-horizon gain has effectively converged to the infinite-horizon LQ gain at that point.

A summary of the metrics across the full grid is presented in Figure 12, with the integral of squared output error (ISE) and the total variation of the control $TV(u)$. The QP solution time stays below 1 ms per call in all combinations, well within the practical budget of 10% of the sampling time (500 ms).

Based on these results, the design is frozen at

$$\boxed{H = 20, \quad R = 10^{-2}}$$

for the remainder of P4. The horizon $H = 20$ is on the ISE plateau and leaves margin for the additional constraints introduced in the following questions. The control weight $R = 10^{-2}$ is small enough to clearly expose input saturation in Q3, yet large enough to produce a smooth steady-state response.

3. Input Bounds: $u \in [0, 100]\%$

By embedding the heater limits (0–100%) directly into the `quadprog` optimization via `lb/ub` arguments, the MPC successfully respects the physical constraints of the hardware. As shown in Figure 13, the bounded controller saturates correctly at 100% during the initial transient, ensuring a realizable control action and avoiding the non-physical 219% peaks of the unconstrained version while maintaining full feasibility (`exitflag` = 1).

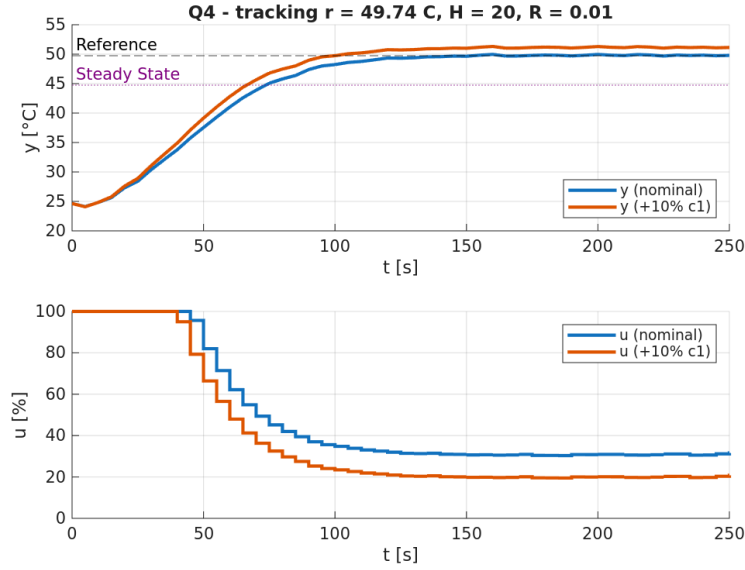


Figure 14: Tracking of $r = 51.32^\circ\text{C}$ with feedforward, with the nominal plant and with a perturbation equivalent to a +12.5% input bias.

4. Tracking with Feedforward and the Mismatch Test

For tracking a non-zero reference $r = y_{ss} + \Delta r$, the controller is reformulated through the change of variables

$$\delta\hat{x} = \Delta\hat{x} - \Delta\bar{x}, \quad \delta\hat{u} = \Delta\hat{u} - \Delta\bar{u}, \quad \delta\hat{y} = \Delta\hat{y} - \Delta r,$$

where the feedforward pair $(\Delta\bar{x}, \Delta\bar{u})$ is the steady-state solution of

$$\Delta\bar{x} = A \Delta\bar{x} + B (\Delta\bar{u} + \hat{d}), \quad \Delta r = C \Delta\bar{x}.$$

In the new variables the problem is again a regulator on $(\delta\hat{x}, \delta\hat{u})$ with the same model and the same cost, and the bounds become $\delta u \in [\Delta u_{\min} - \Delta\bar{u}, \Delta u_{\max} - \Delta\bar{u}]$.

For Q4 there is no disturbance estimator yet, so $\hat{d} = 0$. The reference was chosen as $\Delta r = 5^\circ\text{C}$ (i.e. $r = 51.32^\circ\text{C}$), which gives the feedforward control $\Delta\bar{u} = 5.21$ and $\|\Delta\bar{x}\| = 0.21$.

Choice of the mismatch. The project guide asks for a +10% perturbation of the simulator's constant $c1$. In the implementation,

$$c1 = (I - A) x_{ss} - B u_{ss},$$

with x_{ss} obtained from least-squares fitting. This vector has a norm of the order of 10^{-3} , so a literal 10% increase of $c1$ was also tested but produced an output offset of only 0.11°C , indistinguishable from the measurement noise. Since the objective of this question is to show the steady-state error caused by an unmodelled plant mismatch, the perturbation used in the reported simulations was implemented as an equivalent constant input bias added to the simulator,

$$c1_{\text{pert}} = c1_{\text{nom}} + B d_{\text{real}}, \quad d_{\text{real}} = 0.50 u_{ss} = 12.5\%,$$

which keeps the disturbance in the same input channel that the controller will later estimate with the augmented Kalman filter. Its magnitude is consistent with the DC gain of the identified model ($C(I - A)^{-1}B \approx 0.96^\circ\text{C}/\%$) and produces an open-loop offset of about 1.2°C if uncompensated.

	nominal	perturbed
mean y (last 100 s)	51.26°C	52.63°C
error vs r	-0.06°C	$+1.31^\circ\text{C}$
mean u (last 100 s)	$\sim 35\%$	$\sim 22\%$

Table 3: Tracking performance with and without the input-bias mismatch.

For Figure 14 and Table 3 it can be seen that the nominal case tracks r with essentially zero steady-state error, confirming that the feedforward formulation is correct when the model matches the plant. The perturbed

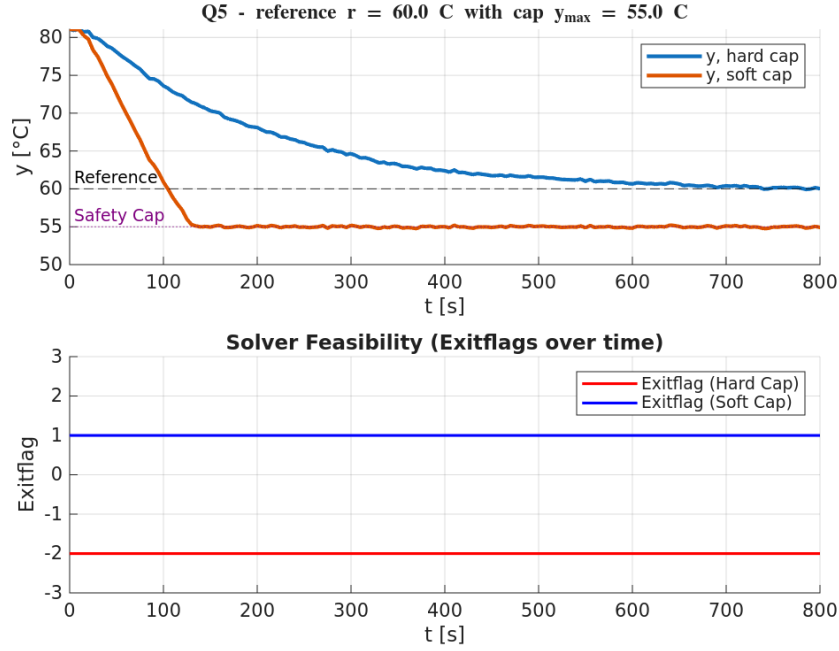


Figure 15: Output safety cap $y \leq 55^\circ\text{C}$ with $r = 60^\circ\text{C}$, starting at $y(0) \approx 89^\circ\text{C}$.

case shows a persistent offset of $+1.31^\circ\text{C}$: the loop reduces u from $\sim 35\%$ to $\sim 22\%$ to counter the unmodelled heat input, but a residual error remains.

Why the offset persists. The MPC's cost is purely quadratic in the shifted variables: when $\delta\hat{x} = 0$ and $\delta\hat{u} = 0$ the cost is zero, so the controller has no remaining gradient to push against. There is no integrator in the loop, and a high-gain proportional regulator on a disturbed plant always leaves an offset equal to the disturbance divided by the loop gain. Removing this offset requires estimating d on-line and injecting it into the feedforward target, which is the role of the disturbance-augmented Kalman filter introduced in the following sections. The perturbation is kept active for Q6/Q7, and removed only for Q5.

5. Output Safety Cap, Hard versus Soft

A safety constraint $y \leq y_{\max} = 55^\circ\text{C}$ was added to the tracking problem, with $r = 60^\circ\text{C}$ deliberately chosen above the cap so that the constraint must bind. In the shifted frame the constraint becomes

$$\delta\hat{y}(i) \leq (y_{\max} - y_{ss}) - \Delta r = -4.98,$$

a negative right-hand side that reflects the gap between the (unreachable) reference and the cap. The mismatch of Q4 is switched off for this question.

The constraint is implemented in two versions:

- **hard:** $\delta\hat{y}(i) \leq y_{\max, \text{shift}}$ enforced exactly. May render the QP infeasible.
- **soft:** $\delta\hat{y}(i) \leq y_{\max, \text{shift}} + \eta_i$, with $\eta_i \geq 0$ and a quadratic penalty $\alpha \eta_i^2$ in the cost, $\alpha = 10^4$. Always feasible.

Forcing infeasibility. With the plant starting at ambient temperature, the hard constraint was never violated by the predicted trajectory and `quadprog` always returned a feasible solution. To trigger the failure mode the question is targeting, the initial condition was set to the equilibrium for $u_{\text{hot}} = 70\%$, i.e. $y(0) \approx 89^\circ\text{C}$ – well above the cap. With $u_{\min} = 0$ (no active cooling) the plant cannot drive its predicted output below 55°C within $H = 20$ samples, and the hard QP becomes infeasible. In this case, `mpc_solve` falls back to $\Delta u = 0$.

As seen in Figure 15 The hard formulation returns `exitflag` = -2 (infeasibility) in every single sample. The fallback $\Delta u = 0$ leaves the absolute control fixed at $u_{ss} + \Delta\bar{u} \approx 39.3\%$, and the plant cools open-loop towards the equilibrium for that input ($\sim 60.5^\circ\text{C}$), violating the cap throughout the experiment.

The soft formulation never fails: the slack variables η_i absorb the unavoidable violation while the heavy penalty $\alpha = 10^4$ prevents abuse. In steady state the output sits on the cap ($\sim 54.99^\circ\text{C}$) and the input drops to $\sim 37\%$ – the solver understands that reaching the reference would violate the cap, and prefers to stay exactly on the boundary.

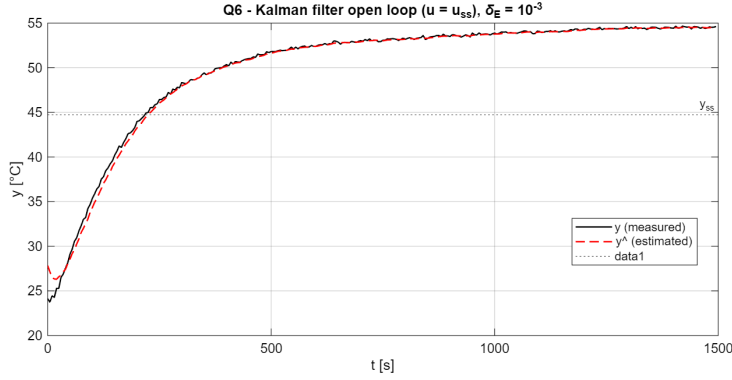


Figure 16: Measured output y and estimated output $\hat{y}$ for $\delta_E = 10^{-3}$, with the filter running in open loop.

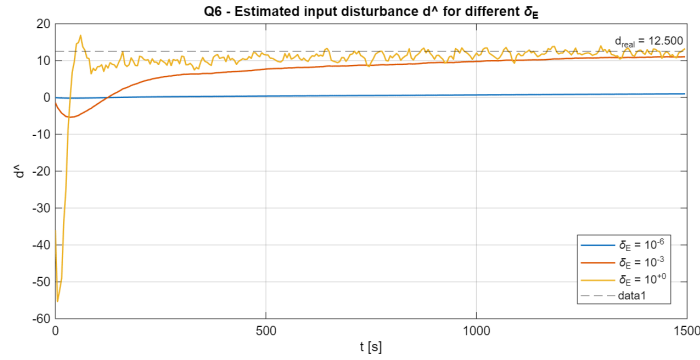


Figure 17: Estimated input disturbance $\hat{d}$ for three values of δ_E . The true value is $d_{\text{real}} = 12.5$ (dashed).

In a real system, even when the hard cap is satisfiable in principle, a disturbance or estimation error can momentarily make the QP infeasible and break the loop. The soft formulation guarantees continued operation at the cost of one extra optimisation variable per stage and is therefore preferred from here on.

6. Augmented Kalman Filter, Open-Loop Test

Up to this point the simulator's state was passed directly to the controller, which is impossible in the real plant where only the output y is measured. A Kalman filter is required to estimate the state from the noisy output sequence. Moreover, Q4 showed that without on-line knowledge of the input bias the MPC leaves a steady-state offset of about 1.31°C . Both problems are addressed at once by augmenting the state with the input disturbance,

$$x_d = \begin{bmatrix} \Delta x \\ d \end{bmatrix}, \quad A_d = \begin{bmatrix} A & B \\ 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} B \\ 0 \end{bmatrix}, \quad C_d = [C \quad 0],$$

with augmented process-noise covariance $Q_{E,d} = \text{blkdiag}(K_e K_e^\top e_{\text{var}}, \delta_E)$ and measurement covariance $R_E = e_{\text{var}}$. The filter gain is computed once as

$$L = \text{dlqe}(A_d, I_{n+1}, C_d, Q_{E,d}, R_E),$$

and, at every sampling instant, the two-step update

$$\hat{x}_d^-(k) = A_d \hat{x}_d(k-1) + B_d \Delta u(k-1), \quad \hat{x}_d(k) = \hat{x}_d^-(k) + L (\Delta y(k) - C_d \hat{x}_d^-(k)),$$

yields the augmented estimate. The first n entries are $\Delta \hat{x}$ (fed to the MPC) and the last entry is $\hat{d}$ (fed to `mpc_steady_state`).

The free parameter is the disturbance-noise variance δ_E : small values make the filter trust the constant-disturbance assumption (slow tracking of d), large values make it react fast at the cost of noise on $\hat{d}$. To choose δ_E , the filter was tested *open loop* ($u = u_{ss}$, no MPC) on the plant with the Q4 perturbation, starting from a deliberately wrong state estimate ($+5^\circ\text{C}$ offset on the output). Three values of δ_E were compared: 10^{-6} , 10^{-3} and 10^0 in Figures 17 and 18.

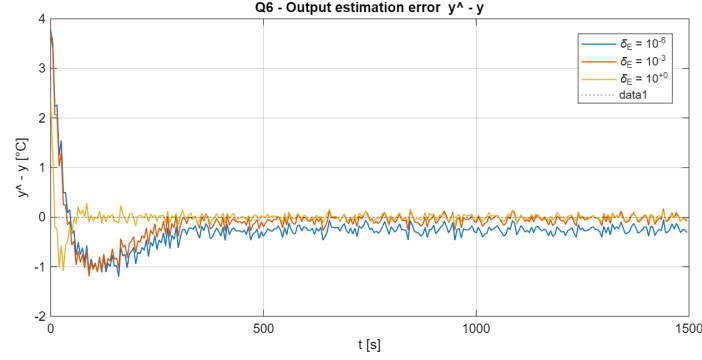


Figure 18: Output estimation error $\hat{y} - y$ for the three values of δ_E .

δ_E	$\hat{d}$ at $t = 1500$ s (real 12.5)	final $\hat{y} - y$	50% settling of $\hat{d}$
10^{-6}	0.98	-0.31°C	— (never)
10^{-3}	11.04	-0.08°C	305 s
10^{+0}	13.25	-0.03°C	40 s

Table 4: Effect of the disturbance variance δ_E on the augmented Kalman estimator.

The initial estimation offset of $+5^\circ\text{C}$ is corrected by the filter within a few samples, after which $\hat{y}$ tracks y tightly.

The values in Table 4 show the expected trade-off: $\delta_E = 10^{-3}$ is already smooth but still slow, while $\delta_E = 10^{+0}$ reacts quickly at the price of a visibly noisier disturbance estimate. For the closed-loop experiment of Q7, an intermediate value between these two behaviours,

$$\delta_E = 10^{-1}$$

was chosen. This value was also checked in the full closed-loop simulation: it converges fast enough to compensate the reference changes while keeping the disturbance estimate smooth. The corresponding gain has norm $\|L\| = 2.715$, with the disturbance component dominating.

7. Full Closed Loop: MPC + Kalman + Tracking

The final P4 experiment combines all previous pieces: MPC with input bounds and soft output cap, augmented Kalman filter ($\delta_E = 10^{-1}$), feedforward target $(\Delta\bar{x}, \Delta\bar{u})$ computed from `mpc_steady_state` using the disturbance estimate $\hat{d}$, applied to the perturbed plant of Q4 ($d_{\text{real}} = 12.5\%$). The reference profile consists of four levels:

$$r \in \{50, 40, 60, 45\}^\circ\text{C},$$

each held for 600 s (2400 s total). The level 60°C is intentionally above the safety cap $y_{\text{max}} = 55^\circ\text{C}$, so the soft cap must intervene during that segment.

To isolate the role of $\hat{d}$, the same simulation was run twice with identical noise:

- **with** $\hat{d}$: the feedforward target uses the Kalman estimate, $(\Delta\bar{x}, \Delta\bar{u}) = \text{mpc_steady_state}(A, B, C, \Delta r, \hat{d})$.
- **without** $\hat{d}$: the filter still runs (same $\Delta\hat{x}$) but $\hat{d}$ is overridden to zero before being fed to `mpc_steady_state`. This is the controller before Q6 was added.

Reference	$\bar{y}$ with $\hat{d}$	offset	$\bar{y}$ without $\hat{d}$	offset
$r = 50^\circ\text{C}$	50.03°C	$+0.03^\circ\text{C}$	51.25°C	$+1.25^\circ\text{C}$
$r = 40^\circ\text{C}$	40.04°C	$+0.04^\circ\text{C}$	41.36°C	$+1.36^\circ\text{C}$
$r = 60^\circ\text{C}$ (cap at 55)	54.93°C	-0.07°C	52.73°C	-2.27°C
$r = 45^\circ\text{C}$	45.01°C	$+0.01^\circ\text{C}$	50.87°C	$+5.87^\circ\text{C}$

Table 5: Steady-state output for each reference segment of the Q7 experiment, computed over the last 40% of the segment.

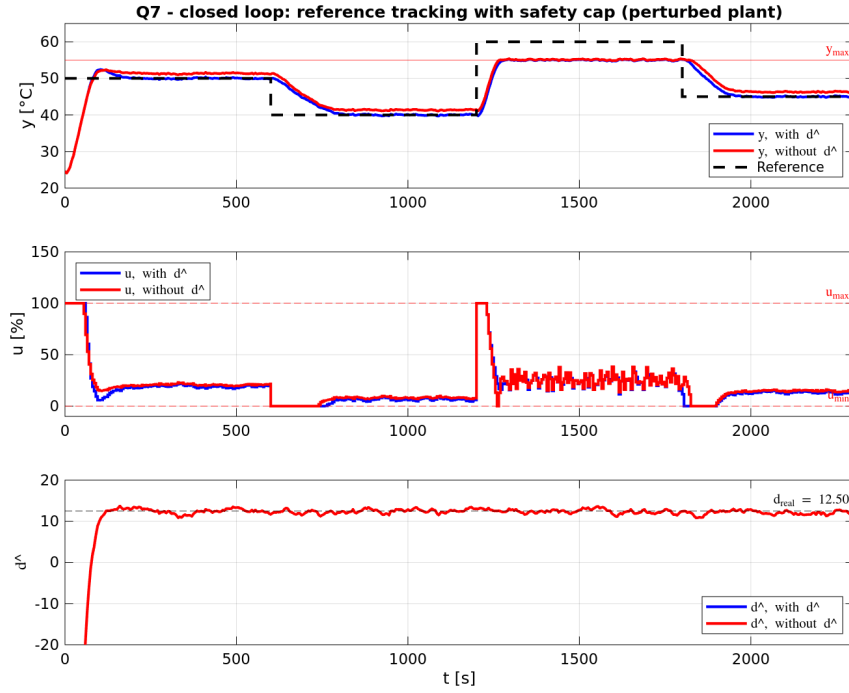


Figure 19: Closed-loop experiment with the full MPC + Kalman scheme on the perturbed plant.

In Figure 19 the MPC’s performance it is observed: with the disturbance estimate tracks every reference level with an offset below 0.1°C , while the configuration without $\hat{d}$ leaves large residual errors and even produces visible chattering during the segments where the cap or the bounds bind. The third segment ($r = 60^\circ\text{C}$) hits the soft cap with $\hat{d}$, settling at 54.9°C ; without $\hat{d}$ the loop oscillates around the cap with u jumping between 0 and 100%. All MPC iterations returned `exitflag` = 1, and the mean QP solve time stayed at ~ 9.7 ms, well within the 500 ms budget. The final value of $\hat{d}$ (~ 12.26) closely matches $d_{\text{real}} = 12.5$, confirming that the filter recovers the unmodelled bias.

Why the disturbance estimate acts as integral action. The augmented model assumes $d(k+1) = d(k)$. The Kalman correction step therefore takes the form

$$\hat{d}(k) = \hat{d}(k-1) + L_d (\Delta y(k) - \hat{\Delta}y(k)),$$

i.e. $\hat{d}$ accumulates (integrates) the innovation $\Delta y - \hat{\Delta}y$. In steady state the innovation is proportional to the output offset, so $\hat{d}$ grows until the offset is cancelled. Feeding $\hat{d}$ into the feedforward target lifts $\Delta\bar{u}$ accordingly, which is exactly the role of the integral term of a PI controller. The advantage of placing the integrator inside the filter (rather than inside the MPC cost) is that the constrained optimisation problem stays purely quadratic and the constraints are still respected as designed.

8. Summary and Conclusions of the Design

The unconstrained regulator was first validated against the analytical receding-horizon solution of P2 (Q1, maximum error $\sim 10^{-11}$). The horizon $H = 20$ and the control weight $R = 10^{-2}$ were then chosen from the $H \times R$ sweep (Q2), giving a controller that is fast enough for the TCLab dynamics while still leaving margin for constraints. Input bounds were added without breaking feasibility (Q3). Tracking with feedforward was implemented through a change of variables, and the persistent offset under model mismatch was demonstrated (Q4), showing why feedforward alone is not sufficient when the plant differs from the model. A safety cap on the output was added in both hard and soft forms, showing that the hard version is fragile while the soft version is always feasible and converges to a sensible boundary behaviour (Q5). A disturbance-augmented Kalman filter was designed and validated open-loop (Q6); $\delta_E = 10^{-1}$ was selected as the compromise between speed and noise on $\hat{d}$. Finally, the full closed loop (Q7) tracked a four-level reference profile with steady-state offsets below 0.1°C , against up to $\sim 5.9^\circ\text{C}$ for the same controller without the disturbance estimate – confirming that the augmented Kalman recovers the integral action that the MPC alone lacks. Overall, P4 completes the simulation design of the constrained MPC, fixes all tuning parameters, and defines the parameters as the baseline to be applied to the real TCLab hardware in P5.

P5 – Application to the real system

In the final phase of the project, MPC and the augmented state Kalman filter were deployed to the physical TCLab plant.

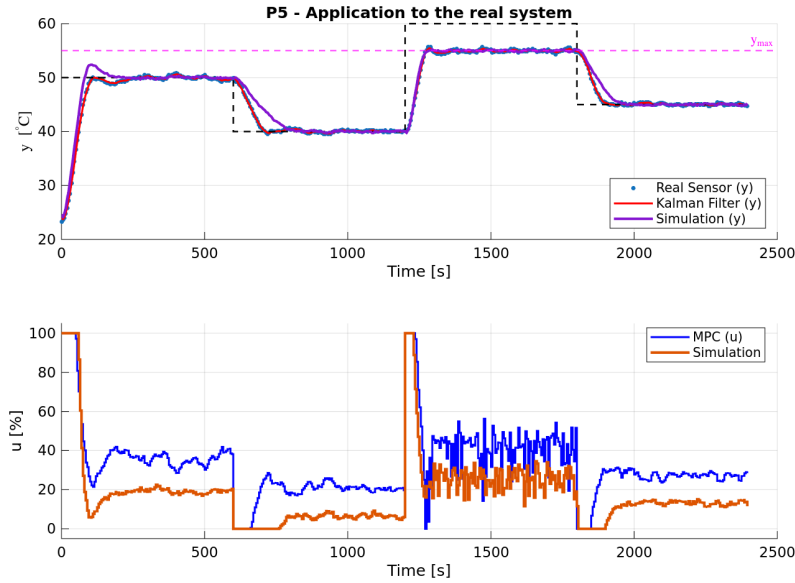


Figure 20: Experimental validation of the MPC and Kalman filter on the TCLab hardware compared to previously simulation results

As illustrated in Figure 20, the MPC achieves zero-offset tracking by accurately reaching the reference set-points without exceeding the safety limit, while the observer effectively tracks the registered temperature. The experimental performance closely aligns with the simulation results, validating the accuracy of the identified linear model and the robustness of the control-estimation architecture in handling real-world plant dynamics. However, a noticeable discrepancy arises regarding the control effort (u): the real system consistently demanded higher steady-state power than the simulation to maintain the same temperature references. This behavior evidences a model mismatch associated with unmodeled convective thermal losses and variations in the actual ambient temperature, which were successfully assimilated and compensated for by the observer.

Finally, during major transient steps, the real temperature response exhibits a more damped behavior compared to the slight overshoot predicted in simulation. This discrepancy is primarily attributed to the artificial $10\%c_1$ perturbation introduced in the simulation, which creates a model mismatch that the controller must overcome. In the real system, the augmented Kalman filter successfully estimates the physical disturbances and unmodeled nonlinearities, effectively providing the necessary compensation to achieve zero-offset tracking. This validation confirms the robustness of the receding-horizon strategy when integrated with real-time disturbance estimation.

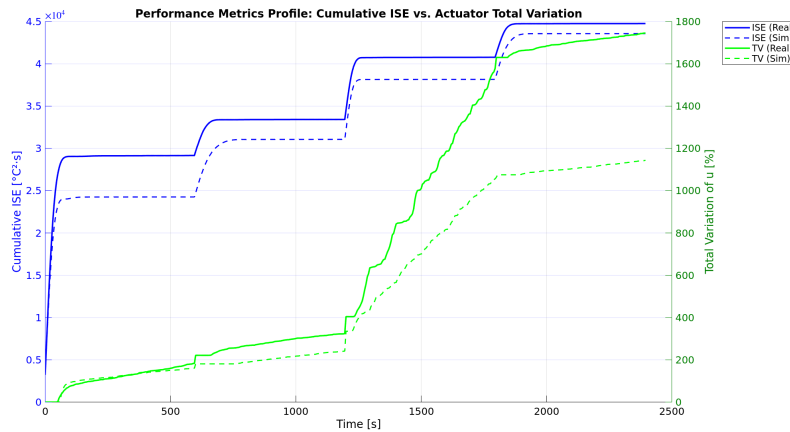


Figure 21: Comparison of cumulative performance metrics (ISE and TV) between experimental and simulated results

These observations are quantitatively corroborated by the cumulative metrics in Figure 21. Regarding the Cumulative ISE (blue curves), both profiles exhibit step-like jumps during transients before flattening into perfectly horizontal asymptotes. This verifies that the augmented Kalman filter enforces zero-offset tracking, successfully canceling the parametric model mismatch. The slight vertical gap between the curves reflects the unmodeled transport delay and non-linear convective losses of the physical plant.

Conversely, the Actuator Total Variation (TV) (green curves) exposes the energetic cost of constraint enforcement. Although both TV profiles match closely for $t < 1200$ s, they severely diverge during the third segment ($1200 \leq t \leq 1800$ s). Here, forcing the system against the 55°C soft safety cap triggers severe high-frequency micro-corrections from the QP optimizer due to sensor noise. Once the reference drops to 45°C and the constraint is deactivated, the real TV slope immediately recovers its low-gradient behavior, demonstrating the localized nature of this actuator stress.

Conclusion

This project allowed the full development of a predictive control framework, starting from constrained optimization and receding-horizon control, followed by model identification and finally the design of an MPC controller with input/output constraints and state estimation.

In the first part, minimization functions were experimented with as the basis of optimization. This knowledge was then applied to a receding horizon control problem, which was compared to the infinite horizon LQR. It was observed that the gains of the receding horizon controller converged to those of the infinite horizon LQR, as the latter represents the optimal solution to the linear control problem without constraints. This differs from the task at hand, hence the decision not to use LQR.

Subsequently, experimental data was collected from the real system, with one dataset used for training and another for validation. After deliberating on the state size, it was determined that a state dimension of 2 yielded good results. By comparing the error with higher-order models, the complexity of the problem was shown to be highly relevant to the computational effort. Thus, the order-2 model was selected.

With the model identified, a sequence of increasingly complex controllers was applied. First, an unconstrained controller was implemented as a baseline. Next, constraints were introduced on the inputs, and the horizon and weighting matrix R were tuned. Following this, a non-zero reference was introduced, and feedforward control was implemented. A Kalman filter observer was then formulated with the plant in open loop to estimate the state and disturbances in the model. Finally, all components were integrated and simulated, demonstrating excellent performance. The validated simulation was then applied to the real system.

The results show that the identified second-order model provides a good compromise between accuracy and complexity, and that the final MPC combined with the augmented Kalman filter is able to track references effectively, reject disturbances, and respect safety constraints. Overall, the work demonstrates the relevance of model-based predictive control for thermal systems with physical limitations and plant uncertainty.

Finally, **AI** tools were utilized mainly to refine the English phrasing and optimize the formatting of the report and images. During the laboratory sessions, AI was also used to help establish a foundational structure for the code; however, all generated content was thoroughly reviewed, frequently corrected, and ultimately validated by the group to ensure technical accuracy.